

OS - Theory - Assignment 01

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BCS-4H

Step 1 — Install dependencies

- Sudo apt update

```
yash-raj@yash-raj-VirtualBox:~$ sudo apt update
[sudo] password for yash-raj:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:8 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:10 http://pk.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:11 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:12 http://pk.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:13 http://pk.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,296 B]
Get:14 http://pk.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:15 http://pk.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:16 http://pk.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 1,055 kB in 2s (534 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
245 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

- Sudo apt upgrade

```
yash-raj@yash-raj-VirtualBox:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  liblvm19
Use 'sudo apt autoremove' to remove it.
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  libzvt-common libcjson1 libpostproc57 libavcodec60
  libgststreamer-plugins-bad1.0-0 libzvt0t64 libavutil50 libswscale7
  libswresample4 libavformat60 libavfilter9
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
  linux-headers-6.17.0-14-generic linux-hwe-6.17.0-14 linux-hwe-6.17.0-14-tools-6.17.0-14 linux-image-6.17.0-14-generic linux-modules-6.17.0-14-generic
  linux-modules-extra-6.17.0-14-generic linux-tools-6.17.0-14-generic
The following packages will be upgraded:
  alsa-ucm-conf apparmor base-files bluez bluez-cups bluez-obexd bsdextrautils bsdutils cloud-init coreutils cpp-13 cpp-13-x86-64-linux-gnu dconf-cli dconf-gsettings-backend
  dconf-service dhcpcd-base distro-info-data eject evolution-data-server evolution-data-server-common fdisk fdup g++-13 g++-13-x86-64-linux-gnu gcc-13 gcc-13-base
  gcc-13-x86-64-linux-gnu gcc-14-base gdm3 gir1.2-gdm-1.0 gir1.2-glib-2.0 gir1.2-gtk-4.0 gir1.2-mutter-14 gir1.2-nm-1.0 gir1.2-packagekitglib-1.0 gir1.2-soup-3.0 gnome-control-center
  gnome-control-center-data gnome-control-center-faces gnome-remote-desktop gnome-settings-daemon-common gnome-shell gnome-shell-common
  gnome-shell-extension-desktop-icons-ng gststreamer1.0-packagekit gststreamer1.0-pipewire inetutils-telnet initramfs-tools initramfs-tools-bin initramfs-tools-core language-pack-en
  language-pack-en-base language-pack-gnome-en language-pack-gnome-en-base ldap-utils libapparmor1 libasan8 libasound2-data libasound2t64 libatomic1 libatopology2t64 libblkid1
  libbluetooth3 libc-bin libc-dev-bin libc-devtools libc6 libc6-dbg libc6-dev libcaml-1.2-64t64 libcc1-0 libcurl3t64-gnutls libcurl4t64 libdconf1 libdjvulibre-text libdjvulibre21
  libebook-1.2-11t64 libebook-1.2-21t64 libebook-contacts-1.2-4t64 libecal-2.0-3 libedata-book-1.2-27t64 libedata-cal-2.0-2t64 libedataserver-1.2-27t64 libedataserverui-1.2-4t64
  libexpat1 libfdisk1 libfprint-2-2 libfprint-2-tod1 libfreerdp-server3-3 libfreerdp3-3 libfwupd2 libgcc-13-dev libgcc-s1 libgdm1 libgfortran5 libglib2.0-0t64 libglib2.0-bin
  libglib2.0-data libgnutls30t64 libgomp1 libgphoto2-6t64 libgphoto2-l10n libgphoto2-port12t64 libgtk-4-1 libgtk-4-bin libgtk-4-common libgtk-4-media-gstreamer libgtop-2.0-11
  libgtop2-common libhwasan0 libinput-bin libinput10 libipa-hbac0t64 libitm1 libldap-common libldap2 liblsan0 libmailcontent-0-0 libnbin-glib4 libnbin-proxy libnbin-utils libmount1
```

- ***sudo apt install build-essential libncurses-dev bison flex libssl-dev libelf-dev dwarves bc wget***

```
yash-raj@yash-raj-VirtualBox:~$ sudo apt install build-essential libncurses-dev bison flex libssl-dev libelf-dev dwarves bc wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
bc is already the newest version (1.07.1-3ubuntu4).
bc set to manually installed.
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libzstd-dev m4 pahole zlib1g-dev
Suggested packages:
  bison-doc ncurses-doc libssl-doc m4-doc
The following NEW packages will be installed:
  bison dwarves libelf-dev libncurses-dev libssl-dev libzstd-dev m4 pahole zlib1g-dev
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 5,389 kB of archives.
After this operation, 23.6 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://pk.archive.ubuntu.com/ubuntu noble/main amd64 m4 amd64 1.4.19-4build1 [244 kB]
Get:2 http://pk.archive.ubuntu.com/ubuntu noble/main amd64 bison amd64 2:3.8.2+dfsg-1build2 [749 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 zlib1g-dev amd64 1:1.3.dfsg-3.1ubuntu2.1 [894 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 libzstd-dev amd64 1.5.5+dfsg2-2build1.1 [364 kB]
Get:5 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 libelf-dev amd64 0.190-1.1ubuntu0.1 [68.5 kB]
Unpacking libzstd-dev:amd64 (1.5.5+dfsg2-2build1.1) ...
Selecting previously unselected package libncurses-dev:amd64.
Preparing to unpack .../5-libncurses-dev_6.4+20240113-1ubuntu2_amd64.deb ...
Unpacking libncurses-dev:amd64 (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package libssl-dev:amd64.
Preparing to unpack .../6-libssl-dev_3.0.13-0ubuntu3.7_amd64.deb ...
Unpacking libssl-dev:amd64 (3.0.13-0ubuntu3.7) ...
Selecting previously unselected package pahole.
Preparing to unpack .../7-pahole_1.25-0ubuntu3_amd64.deb ...
Unpacking pahole (1.25-0ubuntu3) ...
Selecting previously unselected package dwarves.
Preparing to unpack .../8-dwarves_1.25-0ubuntu3_all.deb ...
Unpacking dwarves (1.25-0ubuntu3) ...
Setting up libncurses-dev:amd64 (6.4+20240113-1ubuntu2) ...
Setting up libzstd-dev:amd64 (1.5.5+dfsg2-2build1.1) ...
Setting up m4 (1.4.19-4build1) ...
Setting up libssl-dev:amd64 (3.0.13-0ubuntu3.7) ...
Setting up pahole (1.25-0ubuntu3) ...
Setting up zlib1g-dev:amd64 (1:1.3.dfsg-3.1ubuntu2.1) ...
Setting up bison (2:3.8.2+dfsg-1build2) ...
update-alternatives: using /usr/bin/bison.yacc to provide /usr/bin/yacc (yacc) in auto mode
Setting up libelf-dev:amd64 (0.190-1.1ubuntu0.1) ...
Setting up dwarves (1.25-0ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for install-info (7.1-3build2) ...
yash-raj@yash-raj-VirtualBox:~$
```

Step 2 — Download kernel source

- **Mkdir Kernels**
- **Cd kernels**
- **wget <https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.15.150.tar.xz>**

```
yash-raj@yash-raj-VirtualBox:~$ pwd
/home/yash-raj
yash-raj@yash-raj-VirtualBox:~$ mkdir kernels
yash-raj@yash-raj-VirtualBox:~$ ls
Desktop  Documents  Downloads  kernels  Lab03  'Lab 05'  Music  OS_Lab_24k-0737  Pictures  Public  snap  Templates  Videos  yash.sh
yash-raj@yash-raj-VirtualBox:~$ cd kernels
yash-raj@yash-raj-VirtualBox:~/kernels$ wget https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.15.150.tar.xz
--2026-03-01 01:55:54-- https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.15.150.tar.xz
Resolving cdn.kernel.org (cdn.kernel.org)... 151.101.65.176, 151.101.129.176, 151.101.1.176, ...
Connecting to cdn.kernel.org (cdn.kernel.org)|151.101.65.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 126612456 (121M) [application/x-xz]
Saving to: 'linux-5.15.150.tar.xz'

linux-5.15.150.tar.xz      100%[=====] 120.75M  5.11MB/s   in 23s

2026-03-01 01:56:18 (5.20 MB/s) - 'linux-5.15.150.tar.xz' saved [126612456/126612456]

yash-raj@yash-raj-VirtualBox:~/kernels$
```

- **tar -xf linux-5.15.150.tar.xz** → *Extract the kernel source.*
- **cd linux-5.15.150** → *Enter the extracted kernel source directory.*

```
yash-raj@yash-raj-VirtualBox:~/kernels$ pwd
/home/yash-raj/kernels
yash-raj@yash-raj-VirtualBox:~/kernels$ tar -xf linux-5.15.150.tar.xz
cd linux-5.15.150
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

Step 3 — Copy current Ubuntu config

- **cp /boot/config-\$(uname -r) .config** → *Copies your current kernel configuration to the new source so it will boot correctly.*

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ cp /boot/config-$(uname -r) .config
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

Step 4 — Rename kernel

- **nano Makefile** → Open the kernel Makefile to edit version info.

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ nano Makefile
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

- **EXTRAVERSION = 24k-0737**

```
GNU nano 7.2 Makefile
# SPDX-License-Identifier: GPL-2.0
VERSION = 5
PATCHLEVEL = 15
SUBLEVEL = 150
EXTRAVERSION = 24k-0737
NAME = Trick or Treat
```

Step 5 — Prepare config

- **make oldconfig**

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ make oldconfig
LEX      scripts/kconfig/lexer.lex.c
YACC     scripts/kconfig/parser.tab.[ch]
HOSTCC   scripts/kconfig/lexer.lex.o
HOSTCC   scripts/kconfig/menu.o
HOSTCC   scripts/kconfig/parser.tab.o
HOSTCC   scripts/kconfig/preprocess.o
HOSTCC   scripts/kconfig/symbol.o
HOSTCC   scripts/kconfig/util.o
HOSTLD   scripts/kconfig/conf
.config:806:warning: symbol value 'm' invalid for HAVE_KVM_IRQ_BYPASS
.config:4826:warning: symbol value 'm' invalid for SERIAL_SC16IS7XX_I2C
.config:4827:warning: symbol value 'm' invalid for SERIAL_SC16IS7XX_SPI
.config:5282:warning: symbol value 'm' invalid for GPIO_TPS68470
.config:6544:warning: symbol value 'm' invalid for RADIO_ADAPTERS
.config:8013:warning: symbol value 'm' invalid for SND_SOC_SOF_DEBUG_PROBES
.config:8067:warning: symbol value 'm' invalid for SND_SOC_SOF_HDA_PROBES
.config:11147:warning: symbol value 'm' invalid for ANDROID_BINDER_IPC
.config:11148:warning: symbol value 'm' invalid for ANDROID_BINDERFS
.config:11490:warning: override: SQUASHFS_DECOMP_MULTI changes choice state
```

```
Test firmware loading via userspace interface (TEST_FIRMWARE) [N/m/y/?] n
sysctl test driver (TEST_SYSCTL) [N/m/y/?] n
udelay test driver (TEST_UDELAY) [N/m/y/?] n
Test static keys (TEST_STATIC_KEYS) [N/m/?] n
kmod stress tester (TEST_KMOD) [N/m/?] n
Test memcat_p() helper function (TEST_MEMCAT_P) [N/m/y/?] n
Test livepatching (TEST_LIVEPATCH) [N/m/?] (NEW)
Perform selftest on object aggregation manager (TEST_OBJAGG) [N/m/?] n
Test level of stack variable initialization (TEST_STACKINIT) [N/m/y/?] (NEW)
Test heap/page initialization (TEST_MEMINIT) [N/m/y/?] n
Test HMM (Heterogeneous Memory Management) (TEST_HMM) [N/m/y/?] n
Test freeing pages (TEST_FREE_PAGES) [N/m/y/?] n
Test floating point operations in kernel space (TEST_FPU) [N/m/y/?] n
Test clocksource watchdog in kernel space (TEST_CLOCKSOURCE_WATCHDOG) [N/m/y/?] n
#
# configuration written to .config
#
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

- **Make menuconfig**

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ make menuconfig
UPD      scripts/kconfig/mconf-cfg
HOSTCC   scripts/kconfig/mconf.o
HOSTCC   scripts/kconfig/lxdialog/checklist.o
HOSTCC   scripts/kconfig/lxdialog/inputbox.o
HOSTCC   scripts/kconfig/lxdialog/menubox.o
HOSTCC   scripts/kconfig/lxdialog/textbox.o
HOSTCC   scripts/kconfig/lxdialog/util.o
HOSTCC   scripts/kconfig/lxdialog/yesno.o
HOSTLD   scripts/kconfig/mconf

*** End of the configuration.
*** Execute 'make' to start the build or try 'make help'.

yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

Step 6 — Add Hello World system call

- *Nano kernel/hello.c*

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ nano kernel/hello.c
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

- `SYSCALL_DEFINE0(hello)`

```
GNU nano 7.2
#include <linux/kernel.h>
#include <linux/syscalls.h>

SYSCALL_DEFINE0(hello)
{
    printk("Hello World! I am inside the kernel!\n");
    return 0;
}
```

- *nano kernel/Makefile*

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ nano kernel/Makefile
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

- Add `hello.o` to be compiled as part of the kernel.

GNU nano 7.2

```
# SPDX-License-Identifier: GPL-2.0
```

```
#
```

```
# Makefile for the linux kernel.
```

```
#
```

```
obj-y      = fork.o exec_domain.o panic.o \
            cpu.o exit.o softirq.o resource.o \
            sysctl.o capability.o ptrace.o user.o \
            signal.o sys.o umh.o workqueue.o pid.o task_work.o \
            extable.o params.o \
            kthread.o sys_ni.o nsproxy.o \
            notifier.o ksysfs.o cred.o reboot.o \
            async.o range.o smpboot.o ucount.o regset.o
```

```
obj-y += hello.o
```

```
obj-$(CONFIG_USERMODE_DRIVER) += usermode_driver.o
```

```
obj-$(CONFIG_MODULES) += kmod.o
```

```
obj-$(CONFIG_MULTIUSER) += groups.o
```

- *nano arch/x86/entry/syscalls/syscall_64.tbl* → *Add 548 common hello sys_hello to register the syscall number*

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ nano arch/x86/entry/syscalls/syscall_64.tbl
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

GNU nano 7.2

```
520 x32 execve compat_sys_execve
521 x32 ptrace compat_sys_ptrace
522 x32 rt_sigpending compat_sys_rt_sigpending
523 x32 rt_sigtimedwait compat_sys_rt_sigtimedwait_time64
524 x32 rt_sigqueueinfo compat_sys_rt_sigqueueinfo
525 x32 sigaltstack compat_sys_sigaltstack
526 x32 timer_create compat_sys_timer_create
527 x32 mq_notify compat_sys_mq_notify
528 x32 kexec_load compat_sys_kexec_load
529 x32 waitid compat_sys_waitid
530 x32 set_robust_list compat_sys_set_robust_list
531 x32 get_robust_list compat_sys_get_robust_list
532 x32 vmsplICE sys_vmsplICE
533 x32 move_pages sys_move_pages
534 x32 preadv compat_sys_preadv64
535 x32 pwritev compat_sys_pwritev64
536 x32 rt_tgsigqueueinfo compat_sys_rt_tgsigqueueinfo
537 x32 recvmmsg compat_sys_recvmmsg_time64
538 x32 sendmmsg compat_sys_sendmmsg
539 x32 process_vm_readv sys_process_vm_readv
540 x32 process_vm_writev sys_process_vm_writev
541 x32 setsockopt sys_setsockopt
542 x32 getsockopt sys_getsockopt
543 x32 io_setup compat_sys_io_setup
544 x32 io_submit compat_sys_io_submit
545 x32 execveat compat_sys_execveat
546 x32 preadv2 compat_sys_preadv64v2
547 x32 pwritev2 compat_sys_pwritev64v2
# This is the end of the legacy x32 range. Numbers 548 and above are
# not special and are not to be used for x32-specific syscalls.
548 common hello sys_hello
```


- *nano include/linux/syscalls.h* → *Add prototype asmlinkage long sys_hello(void);*

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ nano include/linux/syscalls.h
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

```
GNU nano 7.2 include/linux/syscalls.h *

    return old;
}

/* for __ARCH_WANT_SYS_IPC */
long ksys_semtimedop(int semid, struct sembuf __user *tsops,
                    unsigned int nsops,
                    const struct __kernel_timespec __user *timeout);
long ksys_semget(key_t key, int nsems, int semflg);
long ksys_old_semctl(int semid, int semnum, int cmd, unsigned long arg);
long ksys_msgget(key_t key, int msgflg);
long ksys_old_msgctl(int msqid, int cmd, struct msqid_ds __user *buf);
long ksys_msgrcv(int msqid, struct msgbuf __user *msgp, size_t msgsz,
                long msgtyp, int msgflg);
long ksys_msgsnd(int msqid, struct msgbuf __user *msgp, size_t msgsz,
                int msgflg);
long ksys_shmget(key_t key, size_t size, int shmflg);
long ksys_shmdt(char __user *shmaddr);
long ksys_old_shmctl(int shmid, int cmd, struct shmid_ds __user *buf);
long compat_ksys_semtimedop(int semid, struct sembuf __user *tsems,
                            unsigned int nsops,
                            const struct old_timespec32 __user *timeout);
long __do_semtimedop(int semid, struct sembuf *tsems, unsigned int nsops,
                    const struct timespec64 *timeout,
                    struct ipc_namespace *ns);

int __sys_getsockopt(int fd, int level, int optname, char __user *optval,
                    int __user *optlen);
int __sys_setsockopt(int fd, int level, int optname, char __user *optval,
                    int optlen);
asmlinkage long sys_hello(void);
#endif
```

Step 7 — Compile the kernel

- `make -j$(nproc)`

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ make -j$(nproc)
```

```
BTF [M] sound/usb/6fire/snd-usb-6fire.ko
LD [M] sound/usb/bcd2000/snd-bcd2000.ko
BTF [M] sound/usb/bcd2000/snd-bcd2000.ko
LD [M] sound/usb/caiaq/snd-usb-caiaq.ko
BTF [M] sound/usb/caiaq/snd-usb-caiaq.ko
LD [M] sound/usb/hiface/snd-usb-hiface.ko
BTF [M] sound/usb/hiface/snd-usb-hiface.ko
LD [M] sound/usb/line6/snd-usb-line6.ko
BTF [M] sound/usb/line6/snd-usb-line6.ko
LD [M] sound/usb/line6/snd-usb-pod.ko
BTF [M] sound/usb/line6/snd-usb-pod.ko
LD [M] sound/usb/line6/snd-usb-podhd.ko
BTF [M] sound/usb/line6/snd-usb-podhd.ko
LD [M] sound/usb/line6/snd-usb-toneport.ko
BTF [M] sound/usb/line6/snd-usb-toneport.ko
LD [M] sound/usb/line6/snd-usb-variak.ko
BTF [M] sound/usb/line6/snd-usb-variak.ko
LD [M] sound/usb/misc/snd-ua101.ko
BTF [M] sound/usb/misc/snd-ua101.ko
LD [M] sound/usb/snd-usb-audio.ko
LD [M] sound/usb/snd-usbmidi-lib.ko
BTF [M] sound/usb/snd-usbmidi-lib.ko
LD [M] sound/usb/usx2y/snd-usb-us122l.ko
BTF [M] sound/usb/snd-usb-audio.ko
BTF [M] sound/usb/usx2y/snd-usb-us122l.ko
LD [M] sound/usb/usx2y/snd-usb-usx2y.ko
BTF [M] sound/usb/usx2y/snd-usb-usx2y.ko
LD [M] sound/virtio/virtio_snd.ko
LD [M] sound/x86/snd-hdmi-lpe-audio.ko
BTF [M] sound/x86/snd-hdmi-lpe-audio.ko
BTF [M] sound/virtio/virtio_snd.ko
LD [M] sound/xen/snd_xen_front.ko
BTF [M] sound/xen/snd_xen_front.ko
LD [M] virt/lib/irqbypass.ko
BTF [M] virt/lib/irqbypass.ko
```

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```


Step 8 — Install kernel

- `sudo make modules_install`

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ sudo make modules_install
```

```
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/usb/line6/snd-usb-variax.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/usb/line6/snd-usb-variax.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/usb/misc/snd-ua101.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/usb/misc/snd-ua101.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/usb/misc/snd-ua101.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/usb/snd-usb-audio.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/usb/snd-usb-audio.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/usb/snd-usb-audio.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/usb/snd-usbmidi-lib.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/usb/snd-usbmidi-lib.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/usb/snd-usbmidi-lib.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/usb/usx2y/snd-usb-us122l.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/usb/usx2y/snd-usb-us122l.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/usb/usx2y/snd-usb-us122l.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/usb/usx2y/snd-usb-usx2y.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/usb/usx2y/snd-usb-usx2y.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/usb/usx2y/snd-usb-usx2y.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/virtio/virtio_snd.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/virtio/virtio_snd.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/virtio/virtio_snd.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/x86/snd-hdmi-lpe-audio.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/x86/snd-hdmi-lpe-audio.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/x86/snd-hdmi-lpe-audio.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/sound/xen/snd_xen_front.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/sound/xen/snd_xen_front.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/sound/xen/snd_xen_front.ko.zst
INSTALL   /lib/modules/5.15.15024k-0737/kernel/virt/lib/irqbypass.ko
SIGN      /lib/modules/5.15.15024k-0737/kernel/virt/lib/irqbypass.ko
ZSTD      /lib/modules/5.15.15024k-0737/kernel/virt/lib/irqbypass.ko.zst
DEPMOD    /lib/modules/5.15.15024k-0737
```

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

- **Sudo make install**

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ sudo make install
[sudo] password for yash-raj:
sh ./arch/x86/boot/install.sh 5.15.15024k-0737 \
    arch/x86/boot/bzImage System.map "/boot"
run-parts: executing /etc/kernel/postinst.d/dkms 5.15.15024k-0737 /boot/vmlinuz-5.15.15024k-0737
* dkms: running auto installation service for kernel 5.15.15024k-0737
* dkms: autoinstall for kernel 5.15.15024k-0737
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 5.15.15024k-0737 /boot/vmlinuz-5.15.15024k-0737
update-initramfs: Generating /boot/initrd.img-5.15.15024k-0737
run-parts: executing /etc/kernel/postinst.d/unattended-upgrades 5.15.15024k-0737 /boot/vmlinuz-5.15.15024k-0737
run-parts: executing /etc/kernel/postinst.d/update-notifier 5.15.15024k-0737 /boot/vmlinuz-5.15.15024k-0737
run-parts: executing /etc/kernel/postinst.d/vboxadd 5.15.15024k-0737 /boot/vmlinuz-5.15.15024k-0737
VirtualBox Guest Additions: Building the modules for kernel 5.15.15024k-0737.

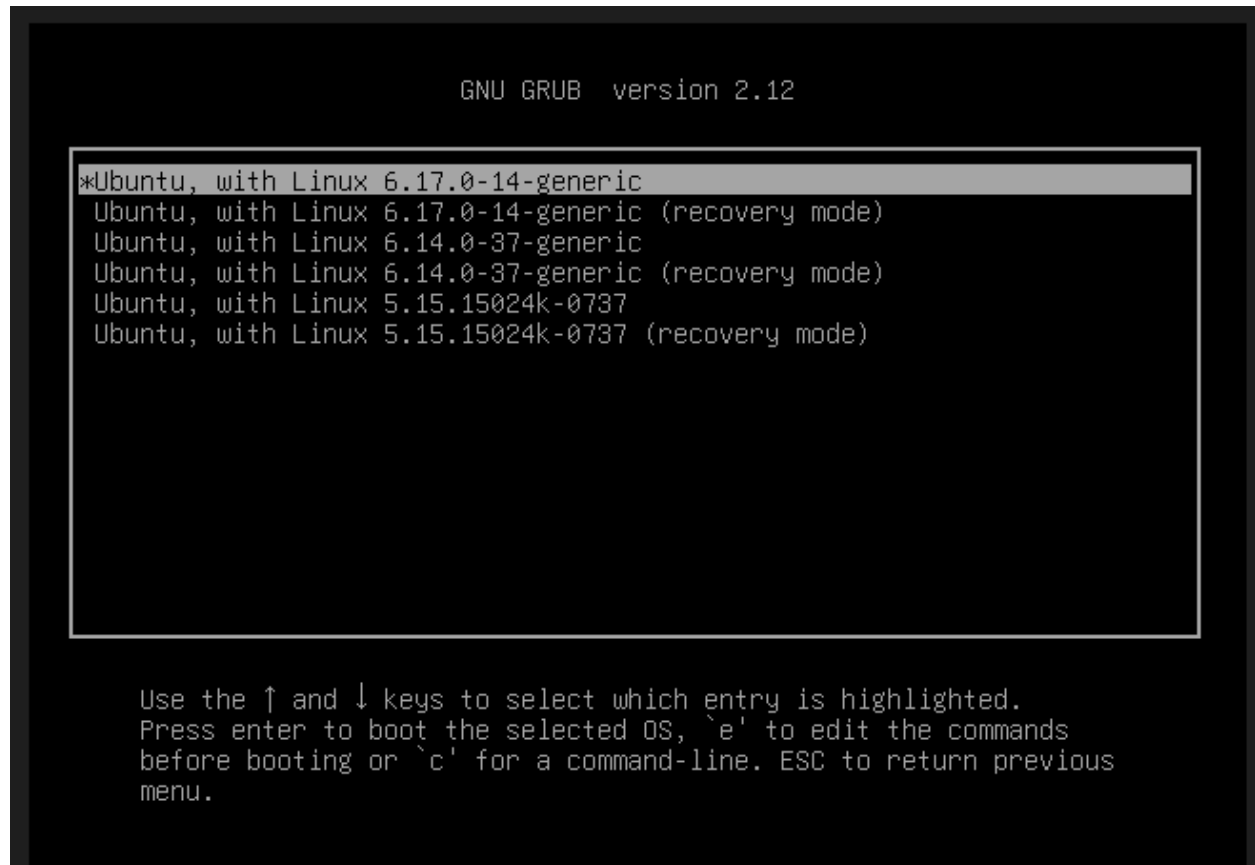
VirtualBox Guest Additions: Look at /var/log/vboxadd-setup.log to find out what
went wrong
run-parts: executing /etc/kernel/postinst.d/xx-update-initrd-links 5.15.15024k-0737 /boot/vmlinuz-5.15.15024k-0737
I: /boot/initrd.img.old is now a symlink to initrd.img-6.17.0-14-generic
I: /boot/initrd.img is now a symlink to initrd.img-5.15.15024k-0737
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 5.15.15024k-0737 /boot/vmlinuz-5.15.15024k-0737
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.17.0-14-generic
Found initrd image: /boot/initrd.img-6.17.0-14-generic
Found linux image: /boot/vmlinuz-6.14.0-37-generic
Found initrd image: /boot/initrd.img-6.14.0-37-generic
Found linux image: /boot/vmlinuz-5.15.15024k-0737
Found initrd image: /boot/initrd.img-5.15.15024k-0737
Found memtest86+x64 image: /boot/memtest86+x64.bin
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

- **Sudo update-grub**

```
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.17.0-14-generic
Found initrd image: /boot/initrd.img-6.17.0-14-generic
Found linux image: /boot/vmlinuz-6.14.0-37-generic
Found initrd image: /boot/initrd.img-6.14.0-37-generic
Found linux image: /boot/vmlinuz-5.15.15024k-0737
Found initrd image: /boot/initrd.img-5.15.15024k-0737
Found memtest86+x64 image: /boot/memtest86+x64.bin
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
yash-raj@yash-raj-VirtualBox:~/kernels/linux-5.15.150$
```

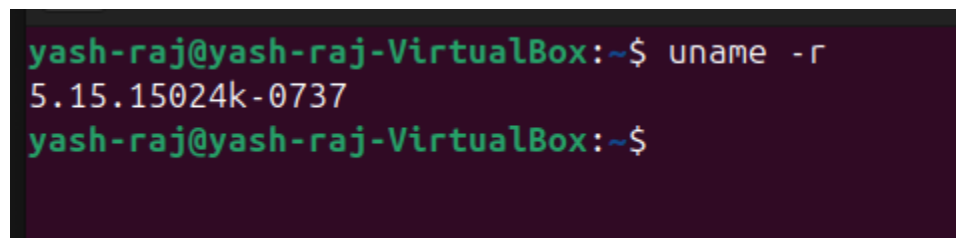
Step 9 — Reboot

- ***Sudo reboot***
- ***ESC***
- ***Select your custom kernel***



Step 10 — Verify

- ***uname -r***



Step 11 — Test Hello System Call

- `nano test.c`

```
yash-raj@yash-raj-VirtualBox:~$ nano test.c
yash-raj@yash-raj-VirtualBox:~$
```

- `syscall(548)`

```
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/syscall.h>

int main()
{
    syscall(548);
    return 0;
}
```

- `gcc test.c -o test`
- `./test`
- `dmesg | tail`

```
yash-raj@yash-raj-VirtualBox:~$ nano test.c
yash-raj@yash-raj-VirtualBox:~$ gcc test.c -o test
yash-raj@yash-raj-VirtualBox:~$ ./test
yash-raj@yash-raj-VirtualBox:~$ sudo dmesg | tail
[sudo] password for yash-raj:
[ 606.766008] audit: type=1400 audit(1772361675.802:187): apparmor="DENIED" operation="open" profile="kernel" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[ 611.854427] audit: type=1400 audit(1772361680.893:188): apparmor="DENIED" operation="open" profile="kernel" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[ 616.863203] audit: type=1400 audit(1772361685.904:189): apparmor="DENIED" operation="open" profile="kernel" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[ 1304.760759] Hello World! I am inside the kernel!
```